Parallel and distributed Computing

Lecture 13

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Today's Topic

GPU Architecture and Programming

Introduction

- A GPU, or a graphics processing unit, is a specialized electronic circuit designed to rapidly process and manipulate memory to accelerate the creation of digital images. Because GPUs have thousands of smaller cores (depending on the model and intended application) compared to CPUs, GPU architecture is optimized for parallel processing. GPUs can handle multiple tasks simultaneously and are faster at graphical and mathematical workloads.

Introduction

- GPU architecture emphasizes throughput via parallelism, while CPU architecture focuses on low-latency sequential execution and flexibility. GPUs specialize in rapid graphical and math operations and have thousands of smaller cores, while CPUs are more generalized with a few larger cores.

Introduction

- Graphics Processing Units (GPUs) are specialized hardware designed to handle parallel computations efficiently. Originally developed for rendering graphics, GPUs are now widely used for **general-purpose computing** (GPGPU) in fields like **machine learning**, **scientific simulations**, and **data analytics**.
- GPUs are designed to perform many computations in parallel, making them ideal for tasks that can be broken down into smaller, independent tasks. Unlike CPUs, which have a few cores optimized for sequential processing, GPUs have thousands of smaller cores optimized for parallel processing.

Key Components of GPU Architecture

1. **Streaming Multiprocessors (SMs)**:
   1. The fundamental processing units in a GPU.
   2. Each SM contains multiple **CUDA cores** (in NVIDIA GPUs) or **stream processors** (in AMD (Advanced Micro Devices) GPUs).
   3. SMs execute threads in parallel and manage resources like registers and shared memory.
2. **CUDA Cores / Stream Processors**:
   1. The basic computation units within an SM.
   2. Each core can execute a single thread.
3. **Global Memory**:
   1. The main memory of the GPU, accessible by all SMs.
   2. High bandwidth but higher latency compared to on-chip memory.

Key Components of GPU Architecture

4.Shared Memory:

1. On-chip memory shared by threads within the same SM.
2. Low latency but limited in size.

5.Registers:

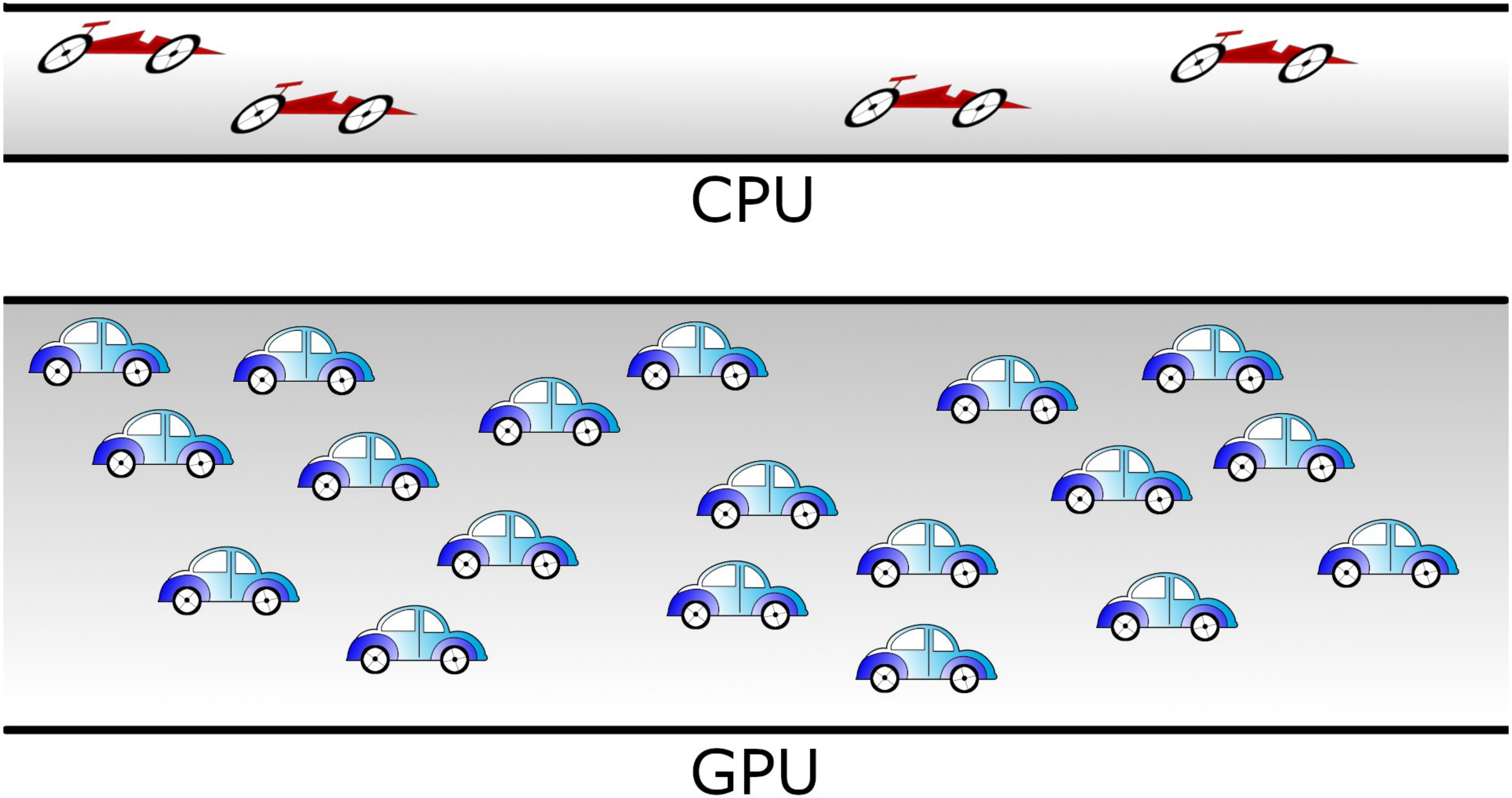
3. Fastest memory available to each thread.
4. Used to store local variables and intermediate results.

6. Cache Hierarchy:

5. Includes L1 and L2 caches to reduce memory access latency.
6. L1 cache is shared within an SM, while L2 cache is shared across SMs.

7. Interconnection Network:

7. Connects SMs, memory, and other components.
8. Ensures efficient data transfer within the GPU.


CPU
GPU

across

- There are many approaches to increase computer processing power like improving the processing power of computer processors, using multiple processors or multiple computers to perform computations or using graphics processing unit (GPU) to speed up computations.
- There are many reasons for parallelization, like speed up execution, overcome memory capacity limit and execute application that is distributed in its nature. The main reason for parallelization is to speed up the execution of applications. Another problem that arises in the era of big data is that the huge data, which need to be processed, do not fit in a single computer memory. Some applications are distributed in their nature, where parts of an application must be located in widely dispersed sites

- **GPU** stands for **graphics processing unit**. GPUs were originally designed specifically to accelerate computer graphics workloads, particularly for 3D graphics. While they are still used for their original purpose of accelerating graphics rendering, GPU parallel computing is now used in a wide range of applications, including graphics and video rendering.

GPU vs. CPU

- GPU parallel computing makes GPUs different from CPUs (central processing units). You can find CPUs in just about every device you own. Built inside just about everything from your Smartphone to your thermostat, they are responsible for processing and executing instructions. Think of them as the brains of your devices.
- CPUs are fast, but they work by quickly executing a series of tasks, which requires a lot of interactivity. This is known as **serial processing**.
- GPU parallel computing is a type of computation in which many calculations or processes are carried out simultaneously. This is known as **parallel processing** and it is what makes GPUs so fast and efficient.
- A CPU consists of four to eight CPU cores, while GPU parallel computing is possible thanks to hundreds of smaller cores.
- GPUs and CPUs work together and GPU parallel computing helps to accelerate some of the CPUs functions.

Advice

Kindly read the book before and after the lecture, if some terms/ words are not covered in the lecture, if appear in exam it might not appear strange or out of course.